THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $2n$	$2(n+5)-10=2n$
2	**(b)** $2n$	$2(n+6)-12=2n$
3	**(d)** $10\times8+9$	Place value
4	**(c)** 121	$11(a+b)$
5	**(d)** 21	Top = a + 2b + c
6	**(c)** 3	$3n+3$
7	**(a)** 9	Divisibility rule
8	**(b)** 52	$4x+16$
9	**(c)** 176	$4x+16$
10	**(c)** $8n$	Expand

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2(n+7)-14=2n$ (2).	2
12	$10a+b$ and $10b+a$ (2).	2
13	$11a+11b=11(a+b)$ (2).	2
14	Second row 8, 9; top 17 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Second row 8, 12 (1); top 20 (1); top $=a+2b+c$ (1).	1+1+1
16	Second row 11, 13 (1); top 24 (1); top $=a+2b+c$ (1).	1+1+1
17	Second row 7, 9 (1); top 16 (1); top $=a+2b+c$ (1).	1+1+1
	OR: Second row 8, 13 (1); top 21 (1); top $=a+2b+c$ (1).	1+1+1
18	Second row 13, 11 (1); top 24 (1); top $=a+2b+c$ (1).	1+1+1
	OR: Second row 9, 8 (1); top 17 (1); top $=a+2b+c$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $3(n+5)-15=3n$, $\div3=n$ (2). (ii) Returns the original number (1). (iii) e.g. $(n+5)\times2-2n=10$ (2).	2+1+2
	OR: (i) $a+b$, $b+c$; $a+2b+c$ (2). (ii) 16 (1). (iii) $14+2b=30$, $b=8$ (2).	2+1+2
20	(i) $x+(x+1)+(x+7)+(x+8)$ (2). (ii) $4x=84$, $x=21$ yes; $4x=114$ no (3).	2+3
	OR: (i) $(s/2)^2-xy=((x-y)/2)^2\geq0$ (3). (ii) 9×9=81 m² (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $4n$	$4(n+3)-12=4n$
2	**(a)** $3n$	$3(n+8)-24=3n$
3	**(d)** $10\times6+4$	Place value
4	**(d)** 20	Top = a + 2b + c
5	**(d)** $2x+5$	Translate
6	**(d)** $2k$, $2k+2$, $2k+4$	Even numbers differ by 2
7	**(b)** a, b, c	Place value
8	**(d)** 64	$4x+16$
9	**(b)** even	$n(n+1)$ is a product of consecutive integers
10	**(a)** commutativity	Property

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$n+(n+1)+(n+2)=3(n+1)$ (2).	2
12	$15\times15=225$ (2).	2
13	$(n+1)^2-n^2=2n+1$ (2).	2
14	$x+(x+1)+(x+7)+(x+8)=4x+16$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dates 4, 5, 11, 12 (1); sum 32 (1); algebra check (1).	1+1+1
16	Dates 7, 8, 14, 15 (1); sum 44 (1); algebra check (1).	1+1+1
17	Dates 9, 10, 16, 17 (1); sum 52 (1); algebra check (1).	1+1+1
	OR: Dates 12, 13, 19, 20 (1); sum 64 (1); algebra check (1).	1+1+1
18	Dates 15, 16, 22, 23 (1); sum 76 (1); algebra check (1).	1+1+1
	OR: Dates 21, 22, 28, 29 (1); sum 100 (1); algebra check (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $A=w(40-2w)$ (2). (ii) $w=10$ (2). (iii) 200 m² (1).	2+2+1
	OR: (i) $99(a-c)$ (3). (ii) $721-127=594=99\times6$ (2).	3+2
20	(i) $5(n+2)$ (3). (ii) 15, 16, 17, 18, 19 (2).	3+2
	OR: (i) $345345=345\times1001$ (2). (ii) $1001=7\times11\times13$ (2). (iii) Another example (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.